A nonregular Edwards descent with nonzero equations

Lean checks a concrete twisted-Edwards point-decomposition input over F25: both descended equations are nonzero quadratics, the affine coordinate algebra has dimension two over F5 and is Artinian, yet their highest homogeneous parts are not regular. Whole gate: 55 declarations, zero failures and zero trusted-native exceptions. Signed standing remains unverified. [pcc:unverified]

Take $K = \mathbb{F}_5[\beta]$, $\beta^2 = 2$, and $a = \beta$, $d = 4$, $R = (1, 1+\beta)$ on $ax^2 + y^2 = 1 + dx^2y^2$. The parameters are valid, the target is smooth and its y-coordinate and a are outside F5. Actual factor-base points $P = (-(2+3\beta), 2)$ and $Q = (2+3\beta, 3)$ add to R with nonzero Edwards denominators. Their explicit images are actual Mathlib points on $Y^2 = X^3 + (2+3\beta)X^2 + (3+2\beta)X$; Lean checks their group sum and R≠O directly. [pcc:unverified]

The exact FGHR2014 printed page 604 polynomial is used, including a/d and the cross coefficient 2(d−a)/d. Universal symmetrization with e1=y1+y2, e2=y1*y2 and actual coefficient extraction in the basis (1,beta) give

$$f_0 = 2(e_1^2 + e_2^2) + 2e_2 + 1, \qquad f_1 = 2(e_1^2 + e_2^2) + 3.$$

Their two highest homogeneous parts coincide and are nonzero. Mathlib `RingTheory.Sequence.IsRegular` is directly disproved for those parts, which is the affine-regularity definition in the paper. [pcc:unverified]

Exact ideal certificates and an actual algebra equivalence prove the following. [pcc:unverified]

$$(f_0, f_1) = (e_1^2, e_2 - 1), \qquad \mathbb{F}_5[e_1, e_2]/(f_0, f_1) \simeq \mathbb{F}_5[T]/(T^2).$$

Thus the affine quotient is finite-dimensional/Artinian and has length 2. Its F5 zero set is exactly {(0,1)}. Multiplicity and support differ. The source's generic degree 4 is not a universal equality. [pcc:unverified]

This disproves the literal universal n = 2 reading of Hypothesis3 with the paper's Edwards representation and nonzero/zero-dimensional affine inputs. It strengthens the prior short-Weierstrass [f,0] witness. A generic-input or n ≥ 3 reading, conditional Theorem 4.1, prime-order sampling distribution and Joux–Vitse Assumption 2 remain separate. No native solver outcome, generic refutation or novelty claim is asserted. [pcc:unverified]

In namespace `EdwardsDescentAudit`, the closed binding is `actual_edwards_descent_counterexample`. Lean 4.35.0 and Mathlib 56a00585 use only propext, Classical.choice and Quot.sound: no holes, custom axioms, native_decide, waivers or raised elaboration limits. All failed attempts and complete gate receipts are retained losslessly. Original FGHR §§1–4/Hypothesis3 and Joux–Vitse pp4–5/7/9–10 are recorded with exact hashes/read scopes. KB unavailable (MCP absent/wrapper127); bounded primary/repository/Mathlib fallback. No paid compute or storage upload. Exact packet/source receipts accompany this report. [pcc:unverified]